

NIZHNEUDINSK, Russian Federation

WMO: 296980

Lat: 54.883N

Lon: 99.033E

Elev: 1350

StdP: 13.99

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.8	-31.1	-40.8	0.6	-36.8	-35.1	0.8	-31.0	16.4	12.2	13.7	11.0	0.3	100	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	84.3	65.7	81.0	64.4	78.0	62.9	68.1	79.8	66.4	77.1	64.8	74.7	4.9	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.3	94.9	72.3	62.6	89.4	71.0	61.0	84.3	69.5	33.3	79.9	31.9	77.3	30.7	75.0	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.8	12.7	10.0	DB	-40.5	90.8	7.8	3.1	-46.0	93.1	-50.6	94.9	-54.9	96.7	-60.6	98.9	(4)
(5)			WB	-40.4	71.6	7.5	2.8	-45.9	73.5	-50.3	75.2	-54.5	76.7	-60.0	78.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.3	-4.8	3.5	20.0	36.5	48.3	60.2	64.7	60.3	47.7	34.0	15.3	0.4		
	DBStd	25.94	13.67	13.32	12.09	8.73	8.42	6.79	4.79	6.01	7.56	9.03	13.39	14.21		
	HDD50	7696	1698	1303	929	410	138	7	0	4	132	497	1040	1538		
	HDD65	12030	2163	1723	1394	855	519	171	64	168	520	960	1490	2003		
	CDD50	1247	0	0	0	6	85	313	455	324	63	1	0	0		
	CDD65	106	0	0	0	0	1	27	55	22	1	0	0	0		
	CDH74	1662	0	0	0	4	120	554	614	334	36	0	0	0		
	CDH80	379	0	0	0	1	20	143	138	72	5	0	0	0		
	WSAvg	3.4	2.2	2.7	3.8	5.0	5.0	3.5	2.9	2.9	3.3	3.8	3.4	2.5		
	PrecAvg	15.9	0.4	0.4	0.3	0.7	1.3	1.8	4.2	3.3	1.9	0.8	0.5	0.4		
(15) Precipitation	PrecMax	21.3	1.3	0.9	1.1	1.9	2.5	3.9	7.1	6.3	3.3	2.0	1.1	0.7		
	PrecMin	10.2	0.1	0.0	0.0	0.1	0.3	0.6	0.7	1.1	0.6	0.0	0.1	0.2		
	PrecStd	3.0	0.3	0.2	0.2	0.5	0.6	0.9	1.6	1.1	0.8	0.6	0.3	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.4	38.4	52.1	71.5	82.8	88.6	88.9	86.4	78.9	64.3	47.0	34.2		
		MCWB	27.9	31.1	40.2	51.3	60.1	65.9	70.1	67.2	61.2	48.5	38.9	29.0		
	2%	DB	26.1	32.5	46.4	63.9	77.7	84.0	83.7	81.5	71.4	57.8	40.3	28.8		
		MCWB	24.0	27.8	36.8	47.1	57.1	64.4	67.0	66.0	57.5	46.7	34.9	25.7		
	5%	DB	20.7	27.8	41.6	58.3	72.4	80.4	80.5	77.8	66.0	51.8	36.4	24.5		
		MCWB	18.5	24.2	34.0	44.5	54.3	63.0	65.7	64.3	54.8	43.3	32.3	22.2		
	10%	DB	15.2	23.2	37.7	53.2	67.2	76.5	77.2	73.7	60.8	46.8	32.7	20.2		
		MCWB	13.7	20.4	31.6	42.0	51.9	61.5	64.2	62.3	52.3	40.5	29.9	18.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.5	31.8	41.2	51.8	62.3	69.6	71.7	70.1	62.8	51.0	39.7	29.5		
		MCDB	30.3	37.3	50.8	69.0	78.4	83.4	83.0	81.6	74.6	61.1	45.4	33.0		
	2%	WB	23.7	27.8	37.6	48.5	58.3	66.5	68.6	67.1	58.7	47.4	35.3	25.8		
		MCDB	26.1	32.2	45.6	60.8	74.9	79.4	79.9	78.6	68.8	56.0	39.4	28.7		
	5%	WB	18.6	24.3	34.5	45.4	55.6	64.7	67.0	65.2	56.0	44.0	32.5	22.2		
		MCDB	20.7	27.8	40.9	57.6	70.0	76.7	77.2	75.8	63.9	51.2	35.6	24.3		
	10%	WB	13.7	20.4	32.0	42.3	52.6	62.6	65.6	63.1	53.4	40.9	30.0	18.5		
		MCDB	15.2	23.1	37.2	52.6	65.5	73.8	74.5	71.8	59.3	46.1	32.8	20.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	20.4	26.6	26.2	22.5	24.5	24.0	20.6	20.5	20.2	18.5	16.2	16.7		
		MCDBR	20.3	27.4	30.0	33.9	36.4	32.9	28.0	29.4	30.7	28.3	18.8	18.7		
		MCWBR	17.8	22.6	21.0	19.2	17.9	14.7	12.6	14.1	16.2	16.9	13.8	16.1		
		MCDBR	20.0	26.4	26.7	31.2	32.5	27.7	23.3	25.9	27.3	26.6	17.7	17.8		
	5% WB	MCWBR	17.7	21.8	18.7	18.4	16.7	13.5	11.5	13.1	14.7	16.3	13.3	15.6		
(40) Clear-Sky Solar Irradiance	taub		0.278	0.313	0.350	0.388	0.419	0.419	0.426	0.403	0.354	0.336	0.308	0.282		
	taud		2.059	2.027	2.045	2.178	2.159	2.260	2.273	2.316	2.405	2.314	2.147	2.017		
	Ebn at Noon		198	233	254	263	261	263	258	257	256	229	185	163		
	Edh at Noon		22	32	40	41	44	40	39	35	28	24	20	18		
(44) All-Sky Solar Radiation	RadAvg		303	619	1072	1433	1624	1837	1665	1402	974	554	291	197		
	RadStd		23	33	57	80	147	118	139	96	94	46	31	20		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page